

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
getwd()
```

```
## [1] "/home/guest/R/RemoteRepositoryClone"
```

```
library(tidyverse)  
library(agricolae)  
library(lubridate)  
library(here)  
#set message = FALSE to have output fit within margins of knitted pdf  
  
lake.raw.data <- read.csv(here(  
  "Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),  
  stringsAsFactors = TRUE)  
lake.raw.data$sampldate <- as.Date(lake.raw.data$sampldate,  
  format = "%m/%d/%y")
```

```
#2
default_theme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        axis.title.x = element_text(color = "black", size = 12),
        axis.title.y = element_text(color = "black", size = 12),
        plot.title = element_text(color = "black", size = 13),
        plot.background = element_rect(fill = "white"),
        panel.background = element_rect(fill = "aliceblue"),
        panel.grid.major = element_line(color = "lightgrey"),
        panel.grid.minor = element_line(color = "grey", linetype = "dotted"),
        legend.text = element_text(color = 'black', size = 9),
        legend.title = element_text(size = 10),
        legend.position = "top")
theme_set(default_theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

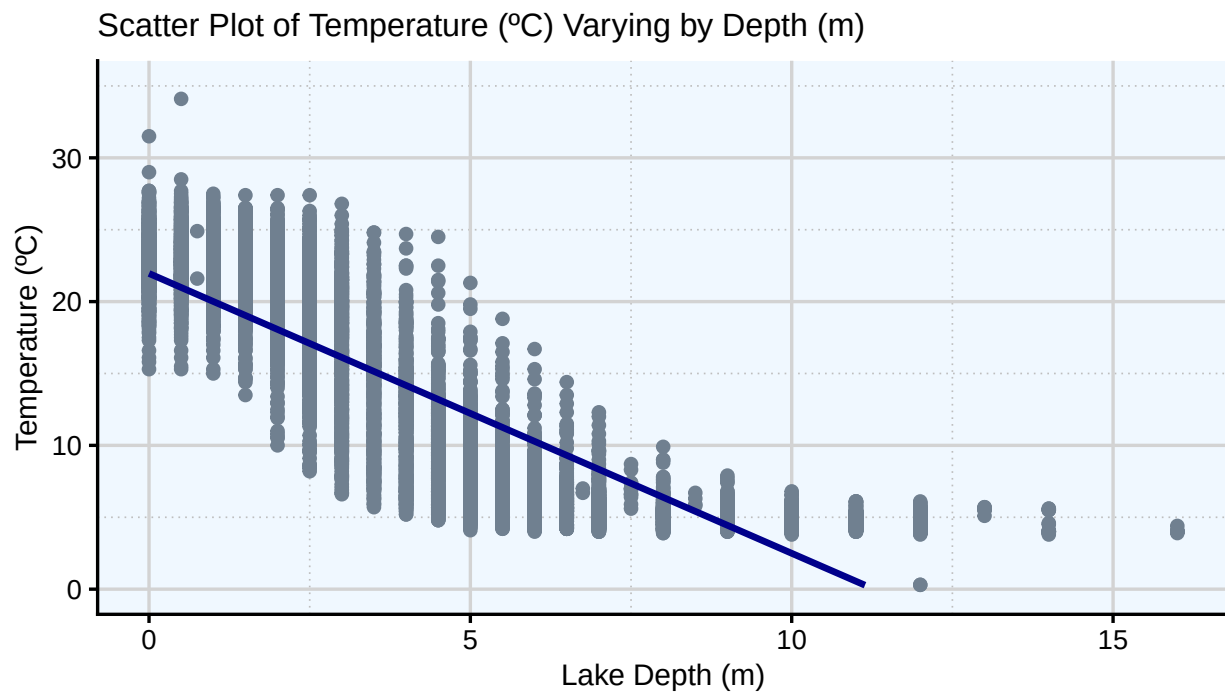
3. State the null and alternative hypotheses for this question: > Answer: H0: Lake temperature recorded during July does not change with depth across all lakes. Ha: Lake temperature recorded during July does change with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
July.data <- lake.raw.data %>%
  filter(month(sampledate) == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()

#5
ggplot(July.data, aes(x = depth, y = temperature_C)) +
  geom_point(color = 'slategrey') +
  geom_smooth(method = 'lm', col = 'darkblue') +
  ylim(0,35)+
  xlab("Lake Depth (m)") +
  ylab("Temperature (°C)") +
  labs(title = "Scatter Plot of Temperature (°C) Varying by Depth (m)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: Looking at the resulting figure, the linear trend suggests that temperature decreases as depth increases. The distribution of points along the linear model show that the linear relationship up until around a 7m depth has a strong linear relationship as this is where the concentration of points is highest around the linear model. The greater the depth, the weaker the linear relationship.

7. Perform a linear regression to test the relationship and display the results.

```
#7
# remember general setup: y = temp, x = depth -> lm(data = '', y~x)
Temp.Depth.regression <- lm(data = July.data, temperature_C~ depth)
summary(Temp.Depth.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = July.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173  -3.0192   0.0633   2.9365  13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 21.95597    0.06792    323.3    <2e-16 ***
## depth      -1.94621    0.01174   -165.8    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: Looking at the regression results using `summary()`, the effectiveness of this model can be established. Looking at R-squared value, depth can account for up to 70% of the variability in temperature, which is a strong dependency. To understand the residual standard error, it is best to also look at the degrees of freedom to determine if the model was accurate or not. The residual standard error is under 4, and given that the best results have low residual standard error, this means the data is pretty accurate because there are 9726 degrees of freedom. This means among over 9000 observations, and data points used in the linear regression model, the residual error is ~3.8. The p-value is less than 0.05 which means the results of this model are statistically significant. And to measure change, the coefficient for depth is -1.95 which is the slope of the linear regression model, so for every 1m of depth change, the temperature decreases by 2°C. Overall, these results show a strong linear relationship between the two variables as depth has a dramatic effect on the temperature.

Multiple Linear Regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
predict_temp_AIC <- lm(data = July.data, temperature_C ~ year4 + daynum + depth)
step(predict_temp_AIC)
```

```
## Start:  AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4      1         101 141788 26070
## - daynum     1         1237 142924 26148
## - depth      1      404475 546161 39189
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.data)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##    -8.57556      0.01134      0.03978     -1.94644

#higher AIC is worse

#10
AIC_improved_regression <- lm(data = July.data, temperature_C~year4 + daynum
                              + depth)
summary(AIC_improved_regression)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: From the AIC results, through the step function as each of the three variables were removed independently, the AIC got larger and the model got worse. This means each variable improved the model, none should be removed, and all are suited to predict temperature. The year, day of data collection, and depth are the final set of explanatory variables that the AIC method suggests should be used to predict temperature. The statistical differences between this model and the previous model are very small. There is a slightly lower RSE indicating a more accurate model and the R-squared value is higher by ~0.1. The p-value is still well below 0.05 so the results remain statistically significant while the degrees of freedom show the observations used in the model changed minimally. This would be a slight improvement over the model that only uses depth as the explanatory variable.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
diff_temp_anova <- aov(data = July.data, temperature_C~lakename)
summary(diff_temp_anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
diff_temp_linear <- lm(data = July.data, temperature_C~lakename)
summary(diff_temp_linear)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = July.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake       -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake      -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake       -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake  -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: Looking at the results of ANOVA test, the p-value is well below 0.05 indicating statistically significant results and showing that the null hypothesis is rejected. This means there is a significant difference in mean temperature among the different lakes. Looking at the linear model, the p-value is the same and confirms that there is significant difference between mean temperatures.

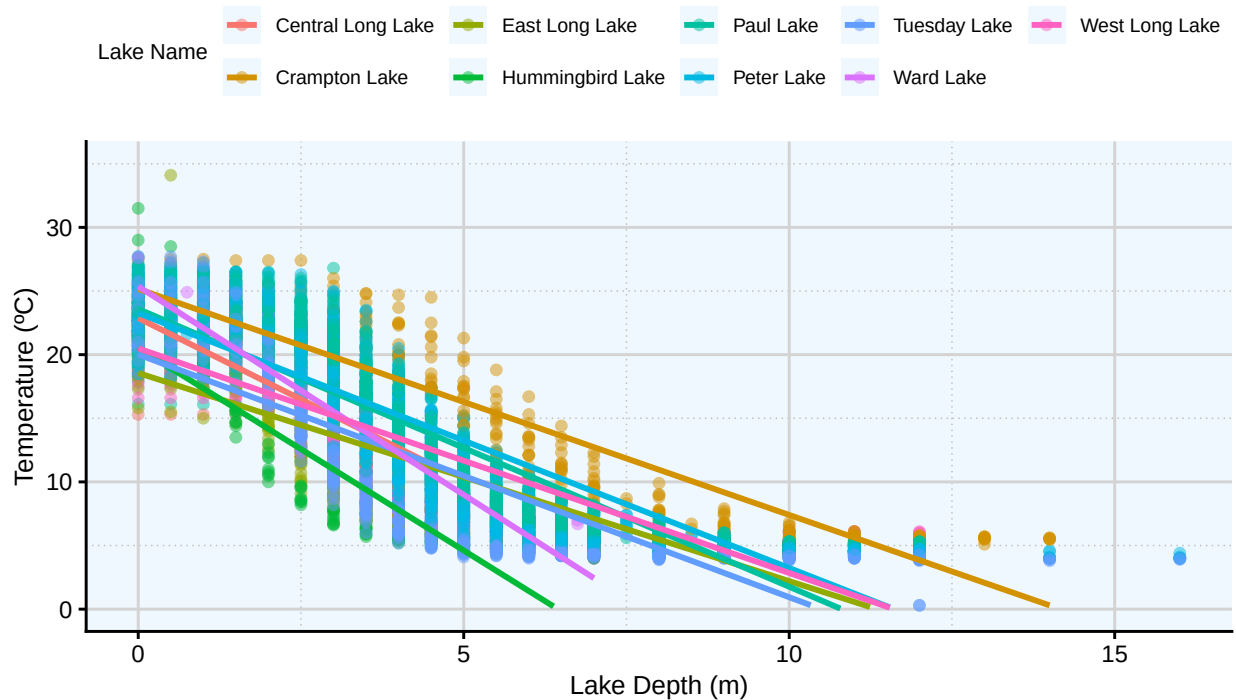
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(July.data, aes(x = depth, y = temperature_C, col=lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = 'lm', se = FALSE, aes(color = lakename)) +
  ylim(0,35)+
  xlab("Lake Depth (m)") +
  ylab("Temperature (°C)") +
  labs(
    title =
      "Scatter Plot of Temperature (°C) Varying by Depth (m) Across Lakes",
    color = "Lake Name")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Scatter Plot of Temperature (°C) Varying by Depth (m) Across Lakes



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(diff_temp_anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = July.data)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
## East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
## Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
## Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
## Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
## Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
## West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
## Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
## Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Looking at the results of the Tukey HSD test, Paul Lake has statistically the same mean temperature as Peter Lake. Between those two there is a difference of -0.498 and a p-value of 0.22 confirming the null hypothesis that there is no change in mean temperature. Ward Lake has an ever greater statistically equal mean temperature as Peter Lake, seen by the p-value of 0.737 which also confirms the null hypothesis between the two lakes. On the other hand, Central Long Lake has statistically the most distinct mean temperature from the other lakes, seen by continuously low p-values, and differences of over 2 for all lake mean temperatures, but a majority

of which are larger than 4. Crampton Lake also has a distinct mean temperature from other lakes, with low p-values but slightly less differences than with Central Long Lake.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: The HSD Test, seen in code as `HSD.test()` would group the lake names in pairwise relationships if their mean temperatures were similar. Looking at Peter and Paul Lake, if this test returned these lakes under the same label, it would show they have the same mean temperature. Another option is to run a two-sample t-test. This will test if the mean temperature between Peter and Paul lake is the same, and if it's true, the null hypothesis is confirmed.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
July.data.wrangled <- July.data %>%
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")
CW_t_test <- t.test(
  July.data.wrangled$temperature_C~July.data.wrangled$lakename)
CW_t_test
```

```
##
## Welch Two Sample t-test
##
## data: July.data.wrangled$temperature_C by July.data.wrangled$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake mean in group Ward Lake
## 15.35189 14.45862
```

```
#looking at linear model data to confirm
CW_t_test_GLM <- lm(
  July.data.wrangled$temperature_C~July.data.wrangled$lakename)
summary(CW_t_test_GLM)
```

```
##
## Call:
## lm(formula = July.data.wrangled$temperature_C ~ July.data.wrangled$lakename)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3519  -7.5286   0.1947   7.0481  13.1414
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      15.3519    0.4087   37.56 <2e-16 ***
## July.data.wrangled$lakenameWard Lake  -0.8933    0.7906   -1.13  0.259
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.289 on 432 degrees of freedom
## Multiple R-squared:  0.002946,    Adjusted R-squared:  0.0006383
## F-statistic: 1.277 on 1 and 432 DF,  p-value: 0.2592
```

Answer: Results of the two-sample T-test determines their mean July temperatures are not different with a p-value above 0.05. The difference between the means is around 0.8, and both the low R-squared value and the high p-value suggest the null hypothesis is confirmed. This matches the results from part 16 where the difference was around 0.89 and a p-value of 0.97 showing they are statistically equal means.